

Digital Access to Justice: Evidence from Women’s Virtual Police Stations in Pakistan*

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This paper studies a large-scale digital reform in Pakistan that changes the *entry point* to justice: Women’s Virtual Police Stations (WVPS), which allow women to report Gender Based Violence (GBV) remotely through a centralized phone-based intake system and digital case management, reducing the need for in-person visits and potentially limiting local discretion at first contact. We combine high-frequency administrative data on police calls and complaints with downstream records on case registration and escalation to examine how digital access reshapes reporting and early-stage justice delivery. Using a difference-in-differences and event-study design around the reform’s rollout, we document three core findings. First, the WVPS increases reported GBV incidents, with effects concentrated in categories where stigma and household control are plausibly most binding. Second, the reform changes the *composition and pathway* of reporting: remote reporting rises while police station walk-in reporting falls, implying substantial substitution in how women engage the police rather than purely mechanical increases in contact. Third, the digital intake produces measurable downstream movement along the justice pipeline, increasing formal registration and follow-up actions with a delay, consistent with capacity constraints and case processing frictions. The effects are larger in places where women’s mobility and social constraints are more severe, indicating that digital access can relax binding constraints on initiating legal claims. The results show that digitizing the first interface between women and the state can expand access to justice in resource-constrained settings, while also shifting where in the pipeline constraints bind.

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1 Introduction

In many low-capacity states, the criminal justice system fails long before a case reaches a courtroom. For women, and especially for women facing gender-based violence (GBV), the most consequential obstacles often arise at the very first step: approaching the police, being taken seriously, and having a complaint recorded in a form that can trigger investigation and prosecution. This initial contact is costly. It can require traveling to a station under household monitoring, recounting intimate experiences to male officers, risking social stigma, and confronting discretionary gatekeeping that screens, downgrades, or refuses cases. As a result, the measured incidence of GBV in official data can reflect not only violence, but also the frictions that prevent women from turning harm into a state-recognized claim (Sukhtankar *et al.*, 2022a; Jassal, 2024).

These frictions matter because justice delivery is a linked, multi-stage process. A case must be initiated, recorded, investigated, charged, and then sustained through adjudication. Discretion and discrimination can compound across these stages, generating large disparities even when each individual decision appears modest in isolation (Jassal, 2024). A growing literature therefore emphasizes reforms that target the *front end* of the system, where many cases are lost. Yet evidence on gender-targeted policing reforms remains mixed. Building on theories of representative bureaucracy and increasing female representation among officers can improve reporting and reduce domestic violence, but effects depend on organizational context and the ability of female officers to operate with meaningful authority (Miller and Segal, 2019a). Creating specialized women’s units or help desks can increase case registration and improve responsiveness, but these reforms may not change the overall rates at which women seek the police if the underlying costs of approaching the state remain high (Sukhtankar *et al.*, 2022a). In parallel, research on policing in public spaces shows that the effectiveness of patrols depends not only on deployment but also on visibility and officer attitudes, highlighting that what citizens *believe* the police will do shapes behavior and outcomes (Amaral *et al.*, 2023).

This paper studies a different policy lever: a digital reform that changes how women access the police in the first place. We examine the rollout of Women’s Virtual Police Stations (WVPS) in Pakistan, which allow women to report incidents remotely through a centralized phone-based intake system integrated with digital case management. Conceptually, the reform targets a core constraint that many institutional reorganizations leave untouched: the need for an in-person encounter at a local station. By enabling remote reporting, the WVPS can reduce mobility and stigma costs, limit exposure to local gatekeeping at first contact, and create a recorded trail that may discipline follow-up. At the same time, digitization can shift rather than eliminate constraints, moving bottlenecks from *initiation* to *processing* if staffing, investigative capacity, or incentives do not adjust.

Our empirical strategy leverages high-frequency administrative data that capture both the *volume* and the *pathway* of reporting. Specifically, we combine call-based reports and complaint intake with downstream records on formal case registration and escalation. We implement difference-in-differences

and event-study designs around the reform’s rollout, comparing GBV-related outcomes to plausible within-system counterfactuals and examining dynamic effects that can distinguish immediate reporting changes from delayed processing and registration responses. This design is well suited to a setting where policy changes can alter multiple margins simultaneously: whether women contact the police at all, whether they contact remotely versus in person, and whether contact translates into formal registration.

Our analysis uses administrative records from two complementary police information systems in Punjab. The Complaint Management System (CMS) records complaints lodged at police stations and tracks whether they are subsequently converted into a First Information Report (FIR), allowing us to separate initial intake from formal legal registration. We merge CMS with Police Rescue 15 (PUCAR), the emergency-call dispatch platform, which captures call-initiated incidents and responses. Together these sources yield a monthly police-station panel that distinguishes (i) walk-in reporting, (ii) call-based reporting, and (iii) downstream formalization into FIRs for domestic violence and other GBV categories. To address strong seasonality in reporting, we construct outcomes as year-over-year differences (same month relative to the previous year) and estimate event-study and difference-in-differences specifications around the WVPS rollout.

The results point to three main conclusions. First, the WVPS increases reported GBV incidents, with effects concentrated in categories where social costs and household constraints plausibly deter in-person reporting. Second, the reform induces substantial substitution in reporting pathways: remote reporting rises while walk-in reporting falls, implying that the intervention changes *how* women access the police, not only whether they do. Third, the increased inflow of digital complaints translates into downstream movement along the justice pipeline with a delay, consistent with processing frictions and capacity constraints.

We see these patterns most clearly in domestic violence, where the WVPS substantially raises call-based reporting while reducing walk-in complaints. Call-initiated domestic-violence reports increase by 23.6 cases per police-station month off a pre-period mean of 11.7, while walk-in complaints fall by 3.9 off a mean of 2.6. The opposing movements point to sizable substitution across reporting technologies, consistent with the WVPS lowering the costs of first contact and bypassing frictions embedded in station-based intake rather than simply expanding enforcement effort. Finally, the higher inflow of complaints translates into downstream formalization with delay. FIR registrations for domestic violence rise by 0.32 per station-month on average relative to a mean of 0.161, and the event-study estimates indicate that FIR effects become positive and statistically significant primarily in later post-period months, consistent with processing constraints and case-handling frictions (Aug–Dec 2024).

This paper contributes to three strands of literature. First, we add to the growing body of work on state capacity and administrative frictions by demonstrating that access to public protection can be constrained at the point of entry into the justice system, rather than at the enforcement stage ([Banerjee et al., 2021](#); [Cheema et al., 2015](#)). Second, we contribute to the literature on gender-responsive policing.

Prior work establishes that female officers and specialized units can increase crime reporting (Miller and Segal, 2019; Perova and Reynolds, 2017; Jassal, 2020); we show that a digital reporting channel can alter the technology of interaction with the police and generate large changes in reporting even absent any expansion of policing resources. Third, we contribute to the literature on digital governance and ICT-enabled public service delivery by providing causal evidence that digital platforms can expand access to justice and bring previously hidden victimization to light.

Within this broader agenda, the paper speaks most directly to an emerging literature asking whether under-reporting of gender-based violence can be addressed by increasing the share of women in the police force (Miller and Segal, 2019) or by creating women police stations (Perova and Reynolds, 2017; Kavanaugh et al., 2019; Jassal, 2020; Bhalotra et al., 2021). We also relate to the literature on GBV deterrence more broadly (Iyengar, 2009; Aizer and Dal Bo, 2009; Amaral et al., 2015; Chin and Cunningham, 2019) and to work showing that increases in women’s political representation produce policy shifts that reflect women’s interests (Beaman et al., 2009; Clots-Figueras, 2011; Iyer et al., 2012; Bhalotra and Clots-Figueras, 2014; Kumar and Prakash, 2017). Most closely related is Iyer et al. (2012), who show in the Indian context that mandating quotas for women in village councils led to a significant increase in GBV reporting. Yet the political economy literature has also documented instances of male backlash against women’s political empowerment (Gagliarducci and Paserman, 2012; Gangadharan et al., 2019), and Jassal (2020) provides compelling evidence that gender bias can permeate the criminal justice system, frustrating the very potential of women police stations. Taken together, these findings underscore the importance of broad empirical investigation into the conditions under which institutional reforms succeed in expanding access to justice for women.

The remainder of the paper proceeds as follows. Section 2 describes the institutional setting and the VWPS program. Section 3 discusses the related literature. Section 4 presents the data and variable construction. Section 5 outlines the empirical strategy. Section 6 reports the main results and heterogeneity analysis. Section 7 concludes.

2 Institutional background: Women’s Virtual Police Station

The Women’s Virtual Police Station (VWPS) is a technology-enabled reporting and support platform implemented by the Punjab Safe Cities Authority (PSCA) in coordination with the Punjab Police. The program was introduced in February 2024 as a digital alternative to traditional police stations with the objective of reducing barriers faced by women in reporting crimes such as domestic violence, harassment, abuse, and cyber-related offenses.

Unlike conventional policing policies that expand police presence or enforcement capacity, the VWPS modifies the mode of interaction between citizens and the police. The program allows women to initiate complaints remotely and confidentially without physically visiting a police station. Victims can access the system through multiple entry points, including the emergency helpline (by dialing “15” and selecting the women’s safety option), a dedicated mobile application, web-based complaint portals,

and real-time communication channels such as live chat and video calls. All contacts are received by trained female communication officers who conduct an initial intake, document the complaint, and assess the nature and urgency of the case.

When a complainant contacts the VWPS, the first step is an intake interview conducted remotely. During this interaction, officers record basic information regarding the incident, including the type of offense, location, identity of the alleged offender (if known), and the immediacy of risk. Based on this assessment, the system classifies the case and assigns a response protocol. Urgent cases—such as ongoing domestic violence or threats to personal safety—are immediately forwarded to the nearest police unit for dispatch. Non-urgent cases enter a follow-up workflow managed by the police station with jurisdiction over the complainant’s location. The VWPS therefore functions as a centralized complaint registration hub that is digitally linked to the existing police infrastructure. Once a complaint is recorded, it is transmitted electronically to the relevant police station, which is responsible for investigation and subsequent procedural steps. Police personnel may contact the complainant, arrange a meeting, collect statements or evidence, or provide referrals to medical, legal, or counseling services. Importantly, the complainant can continue to communicate with officers remotely throughout this process.

A central feature of the program is that complaints initiated through the VWPS can lead to formal criminal registration. When the information provided meets the legal threshold for a cognizable offense, the case may proceed to the registration of a First Information Report (FIR), which formally initiates a criminal investigation under Pakistani law. The VWPS does not itself replace the legal procedures required for FIR registration; rather, it facilitates the steps leading up to them. In practice, police officers guide the complainant through the documentation process and coordinate any required in-person verification, statements, or evidence collection. Consequently, the program primarily lowers the barriers to initiating contact, while the conversion of complaints into FIRs still depends on legal standards and procedural requirements.

Operationally, the VWPS is embedded within the broader PSCA “safe city” infrastructure. The platform integrates with the emergency call system and with digital case-management tools used by the police. Complaints are logged electronically, assigned unique identifiers, and tracked across stages of processing, from initial contact to follow-up actions. This integration allows cases originating from the virtual platform to enter the same administrative workflow as complaints filed in person at a police station. Because the system routes cases to geographically relevant stations, the program does not create a new police jurisdiction. Instead, it serves as an alternative entry point into the existing policing structure. As a result, any downstream actions—investigation, mediation, arrest, or prosecution—remain the responsibility of local police authorities.

The operational design of the VWPS implies a sequential reporting process. The program directly affects the first stage of the interaction with law enforcement: whether victims contact the police at all. By lowering privacy, mobility, and social stigma costs, the virtual channel makes initial disclosure

feasible for victims who would otherwise avoid reporting. However, later stages—particularly FIR registration—require additional procedural steps and often some level of in-person interaction. Therefore, the immediate empirical effect of the policy should be strongest in measures of initial contact (e.g., call-based reporting), while formal legal escalation should occur more gradually and may remain incomplete.

For this reason, the VWPS is best understood as an access-to-justice intervention rather than an increase in enforcement intensity. The program expands the accessibility of the state to potential victims, while leaving the legal and investigative capacity of police institutions largely unchanged. This distinction is central for interpreting the empirical results: observed increases in reporting should be interpreted as improved revelation of victimization rather than necessarily as changes in the underlying incidence of crime.

3 Literature Review

This paper situates the introduction of Virtual Women’s Police Stations (VWPS) at the intersection of three literatures: (i) state capacity and administrative frictions at the citizen–state interface, (ii) gendered barriers to policing and multi-stage justice delivery, and (iii) digital governance and ICT-enabled public service delivery. The common theme is that observed outcomes in administrative systems reflect not only underlying incidence, but also the architecture through which claims are submitted, processed, and formalized. By shifting the locus of complaint registration from a physical to a digital node, the VWPS intervention tests whether technology can overcome the structural and social barriers that traditionally inhibit access to justice.

3.1 State capacity, administrative frictions, and the entry node

First, we build upon the literature examining the binding constraints to public service delivery in South Asia. A central lesson from work on state capacity in low- and middle-income settings is that public services often fail at the interface between citizens and frontline agents. Discretion, low effort, and weak accountability frequently generate large implicit “taxes” on access, especially for citizens with limited bargaining power [Banerjee *et al.* \(2021\)](#); [Cheema *et al.* \(2014\)](#). In the context of policing, these frictions are particularly salient because reporting is inherently a disclosure choice: citizens must incur costs to initiate contact, and frontline officers can discourage, downgrade, or delay complaints before they become officially recorded.

Much of the policy response in this space has attempted to change frontline behavior through training, incentives, or organizational reforms. Yet evidence suggests that shifting police behavior inside the existing station architecture can be difficult. For example, large-scale reforms in India show limited long-run effects of “soft skills” training on police–citizen interactions [Banerjee *et al.* \(2021\)](#)

This motivates an alternative approach such as structural interventions, that alters the reporting channel itself. VWPS operationalizes this “bypass” mechanism by centralizing intake and allow-

ing victims to initiate contact remotely, potentially reducing exposure to idiosyncratic station-level gatekeeping while leaving enforcement capacity largely unchanged.

3.2 Gender Representation, Reporting Costs and Justice Pipeline

Second, our work relates to the extensive literature on the determinants of reporting Gender-Based Violence (GBV). A central insight from this domain is that the under-reporting of crimes against women is endogenous to the composition of the police force. [Miller and Segal \(2019b\)](#) provide robust evidence from the US that increasing the share of female officers significantly raises the reporting of violent crimes against women. The mechanism they propose is not merely one of incidence, but of reception: female victims perceive female officers as more approachable, lowering the psychological and reputational costs of engagement. This is supported by qualitative evidence from [Mello \(2019\)](#), who documents how male-dominated policing environments create “procedural unfairness” that discourages women from initiating legal action.

In the South Asian context, [Iyer *et al.* \(2012\)](#) find that mandated female political representation led to a sharp increase in *documented* crimes against women. Crucially, they attribute this to a reduction in reporting costs rather than an increase in victimization, validating the “reporting channel” hypothesis. Similarly, social norms play a governing role; [Acemoglu and Jackson \(2017\)](#) model how laws often go unenforced when they conflict with prevailing social norms, as citizens refuse to act as “whistleblowers.” [Jensen and Oster \(2009\)](#) further show that exposure to external information sources (e.g., cable television) can alter these norms and reduce the acceptability of domestic violence. Our paper tests whether digital access acts similarly to political representation or media exposure by lowering the threshold for “whistleblowing” against intimate partner violence.

A core implication of this literature is that reforms can increase recorded GBV without increasing underlying violence. The relevant welfare object is therefore not “crime counts” per se, but movement along the justice pipeline: does a report become a legally active case and trigger protection? Recent work shows that gender disadvantage can accumulate at multiple stages—police response, judicial outcomes, and the conversion of complaints into cases—so reforms that only shift the entry margin may not deliver legal remedies ([Jassal, 2024](#)). This multi-stage perspective is central for interpreting WVPS, because the program is designed to relax the first bottleneck (contact), while the conversion step (e.g., FIR registration) remains mediated by legal thresholds and local station incentives.

3.3 Physical vs. Virtual Interventions

The most direct policy benchmarks for WVPS are reforms that create women-facing institutional capacity within policing. Evidence from Brazil suggests that Women’s Police Stations can reduce intimate partner violence and related harms, consistent with increased reporting and improved protection [Perova and Reynolds \(2017\)](#). In India, women police stations and related institutional reforms increase reporting of violence against women, and women-facing entry points within stations (such as women’s

help desks) can increase FIR registration [Amaral *et al.* \(2021\)](#). Related work asks whether under-reporting can be addressed by increasing women’s representation within law enforcement ([Miller and Segal, 2019a](#)) or by expanding access to women-specific policing institutions ([Perova and Reynolds, 2017](#); [Kavanaugh *et al.*, 2019](#); [Amaral *et al.*, 2021](#)).

Unlike [Amaral *et al.* \(2021\)](#) and [Sukhtankar *et al.* \(2022b\)](#), who study interventions that still require physical travel, a non-trivial cost in Pakistan. We examine an intervention that brings the state to the citizen. This paper extends that agenda by moving from “women-facing capacity inside the station” to “women-facing entry outside the station.” Physical stations and in-station help desks can be effective but may be costly to scale and may not eliminate the mobility and stigma costs borne by victims who must appear in person. WVPS is a lower-infrastructure alternative that changes the reporting technology itself, allowing us to test whether lowering entry costs yields (i) more contact, (ii) substitution across reporting channels, and (iii) downstream formalization in the form of FIR registration.

We also relate to the broader literature on deterrence and policy responses to GBV, which studies how enforcement and related policies affect incidence and reporting ([Iyer *et al.*, 2012](#); [Aizer and Dal Bo, 2009](#); [Chin and Cunningham, 2019](#)). A recurring empirical challenge in this literature is that measured GBV is jointly determined by violence and reporting. Because WVPS is designed to reduce reporting costs rather than intensify enforcement, it provides a setting in which increases in recorded GBV can be interpreted primarily as improved revelation and access, and where the formalization margin (FIRs) offers an additional welfare-relevant outcome.

3.4 Digital Governance and Monitoring

Finally, we contribute to the literature on digital governance and monitoring. Technological interventions have been shown to improve state capacity by lowering barriers to entry and enhancing top-down monitoring. [Lewis-Faupel *et al.* \(2016\)](#) find that electronic procurement in India and Indonesia improved infrastructure quality by reducing friction for bidders, effectively democratising access to government contracts.

In the specific context of Punjab, Pakistan, [Callen *et al.* \(2020\)](#) demonstrate that smartphone-based monitoring significantly increased the inspection rates of government health officials. This finding is critical for our design, as it establishes that digital tools can successfully monitor and incentivise agent behaviour in this specific bureaucratic setting. Furthermore, [Olken \(2007\)](#) distinguishes between “grassroots” and “top-down” monitoring, finding the latter more effective in reducing corruption. The WVPS functions as a hybrid: it relies on grassroots initiation (the victim calls) but utilises top-down digital processing to prevent local suppression of cases.

By estimating the differential impact of this digital channel on urban versus rural stations, we also contribute to the broader literature on the elasticity of crime with respect to enforcement [Draca *et al.* \(2011\)](#); [Blattman *et al.* \(2017\)](#), providing new evidence on whether digital platforms can bridge the gap

in access to justice where physical state presence is weak.

3.5 Political economy, social norms, and backlash

A final strand of research highlights that reforms expanding women's voice can induce policy changes in women's interests but can also provoke backlash. Women's political representation shifts policy and public goods in ways that reflect women's preferences and constraints (Hessami and Da Fonseca, 2020; ?; Clots-Figueras *et al.*, 2017; Kumar and Prakash, 2017). At the same time, political economy research documents contexts in which women's empowerment triggers adverse behavioral responses or strategic resistance by men (Gangadharan *et al.*, 2019). In the justice context, evidence of gender bias inside criminal justice institutions implies that women-facing policing institutions may not deliver their intended benefits uniformly across locations (Jassal, 2020, 2024).

This perspective motivates our heterogeneity analysis: if local norms govern the costs of “coming forward” and the social penalties attached to formal legal action, WVPS may have large effects on initial contact precisely where women are most constrained, but weaker effects on formalization where proceeding to an FIR is socially costly. The paper therefore treats social environment not as a peripheral correlation but as a core modifier of how access reforms translate into legal outcomes.

4 Data

This study uses administrative data on reported cases handled by police stations in the context of the Virtual Women Police Station (VWPS) program. The primary sources of case data are the Complaint Management System (CMS) and the Police Rescue 15 (PUCAR 15) system, which are institutional data platforms operated by law enforcement agencies in Pakistan.

The CMS is a digital complaint tracking and case registration platform implemented by the Punjab Police, designed to facilitate the lodging, categorization, assignment, and tracking of complaints by members of the public. CMS enables law enforcement to record complaints systematically, assign them to the appropriate police station, and monitor case progression through unique reference identifiers, improving transparency and oversight over case intake and management. It captures key administrative details such as the date of the complaint, the reporting channel, and the police station responsible for follow-up, thereby creating a structured record of criminal complaints over time.

A key advantage of CMS is that it allows us to identify which recorded complaints were subsequently formalized as First Information Reports (FIRs)—the standard legal registration of a criminal case—enabling us to distinguish (i) initial reporting/complaint intake from (ii) escalation into formal criminal proceedings.

The PUCAR 15 system is a centralized emergency response and case logging platform used by police to record and manage incoming calls and dispatched responses across regions. It operates as a real-time case intake system for emergency calls and dispatches, providing additional administrative detail on events that require a police response. The structure of PUCAR 15 data complements CMS

records by documenting incidents captured through the emergency response channel, offering a broader view of police service demand and case flows at the station level.

Together, the CMS and PUCAR 15 datasets constitute a panel of police station–period observations, with outcomes that include counts of reported cases, incident types, and reporting channels. This panel structure facilitates a temporal analysis of case reporting patterns before and after the implementation of the VWPS initiative. By linking case counts to specific police stations and time periods, we can examine how the introduction of virtual reporting mechanisms influenced both the overall volume and composition of reported cases — particularly those involving gender-based violence.

Our outcomes focus on reporting related to domestic violence against women, constructed as counts per police station-month: Number of reported cases classified as domestic violence reported by walk-in to the police station (CMS) or by emergency calls (PUCAR). As well as the domestic violence cases converted to FIR. As additional variables, we analyzed all cases against women categorized in Table A1 and a placebo variable that includes all non-gender cases reported to the police station, excluding losses and miscellaneous categories.

The administrative records span 2021–2024. Table 1 presented the principal statistics of this period. The average number of cases by police station for all reports are 23,446 and reported by emergency calls are 9,263 cases. Looking in the domestic violence cases, the average number of reported cases are 1,135, and 970 of this cases are reported by 15 calls.

Table 1: Descriptive statistics

	All reports (1)	15 calls (2)
Average number cases	23,446	9,263
Average Domestic Violence cases	1,135	970
Total number of cases	19,413,512	7,669,852

Notes: Reported cases at police station level between 2021 and 2024. Column (1) of all reports include reported cases by walk-in the police station and by 15 calls, Column (2) only include the 15 calls reported cases.

To address strong monthly seasonality in reporting, we construct “excess cases” as changes relative to the same month in the previous year. Using the 2021–2024 series, for each police station i and month t we created the differenced variables as follows:

$$Y_{i,t} = \tilde{Y}_{i,t} - \tilde{Y}_{i,t-12} \quad (1)$$

where $\tilde{Y}_{i,t}$ is the number of police station i cases in month t ; $\tilde{Y}_{i,t-12}$ is the number of police station i cases in the same month the previous year $t - 12$; and $Y_{i,t}$ is the differenced variable that will be used in all the estimations.

This transformation removes station-specific seasonal patterns common to the same calendar month and helps isolate program-related changes.

The treatment definition is motivated by the operational mechanism of the VWPS. The program primarily operates by facilitating remote reporting and lowering access barriers for victims who are unable or unwilling to physically visit a police station. Consequently, police stations that historically received a larger volume of call-initiated reports were more likely to experience a meaningful increase in reporting once the virtual reporting channel became available. The treatment group is defined based on police stations' baseline exposure to remote reporting channels prior to the implementation of the VWPS program. Specifically, we classify as treated the 30% of police stations with the highest number of cases reported through emergency calls (PUCAR) in the pre-program period (from 2021 until the implementation of the program).

This approach captures heterogeneous exposure to the reporting technology introduced by the program. Stations with high baseline call volumes represent areas where victims relied more heavily on non-in-person interaction with the police, suggesting the presence of latent reporting demand constrained by access barriers. The VWPS program relaxes these barriers by allowing confidential reporting from home and enabling the initiation of formal procedures without a station visit. Consequently, the program should disproportionately affect reporting behavior in these locations.

Importantly, the treatment is therefore not defined by administrative targeting but by differential potential usage of the reporting channel. The empirical strategy exploits this variation in baseline remote-contact demand to identify the effect of the VWPS on reporting behavior.

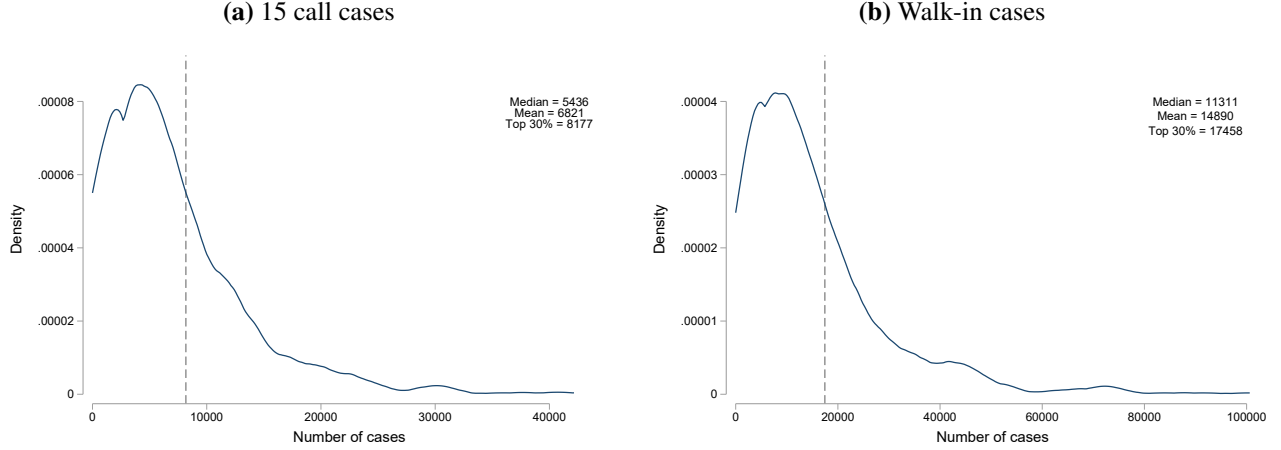
We construct an alternative exposure measure based on in-person reporting activity, defining treatment as the top 30% stations in the number of complaints filed directly at the police station using CMS records. This allows us to test whether results depend specifically on call-based exposure or on general reporting intensity.

Figure ?? displays the distribution of the total number of cases by police station, separately for emergency call reports (panel a) and walk-in reports filed directly at the station (panel b). For call-based reporting, the average station records approximately 7,000 cases during the period, while stations in the top 30% handle roughly 8,000 or more cases. In contrast, walk-in reporting is substantially higher: the mean number of cases is about 15,000, and stations in the top 30% exceed 15,000 reported cases.

Although the administrative records span 2021–2024 the analysis of program effects begins in July 2023. Around this date, the police introduced a new reporting and case-management policy that modified the procedures through which complaints were recorded and processed. This change affects the comparability of case counts in earlier periods, as observed reporting patterns reflect both behavioral responses and administrative practices.

Because the exact operational details of the reform are not consistently documented across stations, we treat July 2023 as a structural break in the measurement process and restrict the estimation sample to the post-reform period. This restriction ensures that variation in reported cases is driven by behavioral changes in reporting rather than by shifts in police recording practices. Under this approach, changes in reporting are more plausibly attributable to the VWPS program instead of contemporaneous

Figure 1: Total cases distribution



Note: Distribution function of the total number of cases reported in each police station by 15 calls and walk-in cases, between 2021 and 2024.

modifications in the police reporting system.

5 Methodology

We estimate the effect of the Virtual Women Police Station (VWPS) using an event-study difference-in-differences design in a balanced panel of police stations observed monthly. Our main analysis uses a balanced panel at the police-station $\times$ month level with one observation per station-month for July 2023–December 2024. The outcomes are constructed as year-over-year differences (same month relative to the previous year) to flexibly remove month-specific seasonality in reporting.

The VWPS reform expands a remote entry point into the justice system. Our identification strategy exploits heterogeneous exposure to this reform across police stations, proxied by baseline reporting intensity. Specifically, we define a time-invariant treatment indicator for stations that lie in the top 30% of pre-program reporting volume by emergency calls (with the baseline measured over the pre-period). This approach compares the evolution of outcomes in “high-exposure” stations to “lower-exposure” stations before vs. after the VWPS implementation.

We estimate a dynamic difference-in-differences event study, which traces the evolution of outcomes relative to the implementation month. To absorb all station-invariant differences and common month shocks, our baseline specification includes police-station fixed effects and calendar month fixed effects:

$$Y_{it} = \alpha_i + \gamma_t + \sum_{k \neq -1} \beta_k (Treated_i \times 1[Event_time_t = k]) + \varepsilon_{it} \quad (2)$$

where α_i are police-station fixed effects and γ_t are month fixed effects capturing common shocks and aggregate reporting changes. The event-time indicators $1[Event_time_t = k]$ index months relative

to the VWPS introduction (with $k = 1$ as the omitted pre-period reference month). The coefficients β_k trace the full the dynamic evolution of program effects : leads ($k < 0$) test for pre-trends and lags ($k \geq 0$) estimate treatment effects over time. The base period is January 2024, because the program started its implementation in February 2024.

The dependent variables Y_{it} are the differenced outcomes of domestic violence reports against women, women-related cases and FIR-converted cases. The treatment variable $Treated_i$ equals 1 for stations in the top 30% of baseline reporting intensity and 0 otherwise. Standard errors are clustered at the police-station level to allow arbitrary serial correlation within stations over time.

5.1 Interpretation and assumptions

The design relies on a parallel trends assumption in the absence of the VWPS: conditional on station fixed effects and common month shocks, treated and control stations would have exhibited similar trends in year-over-year changes in reporting. We assess this assumption in two complementary ways.

First, our event-study design relies on the standard parallel trends assumption: absent the VWPS, the outcomes in treated and control stations would have evolved similarly over time (conditional on station and month fixed effects). We assess this assumption using (i) visual inspection of pre-treatment coefficients and (ii) joint tests that pre-treatment coefficients are zero.

Second, to show that conditional on station fixed effects and common month shocks, treated and control stations would have exhibited similar trends in year-over-year changes in reporting. We estimate the same event-study model using non-gender crime categories (reported to the police station, excluding losses and miscellaneous categories) as a placebo: systematic post-period changes in placebo outcomes would indicate a broader reporting/recording shift unrelated to VWPS.

5.2 Robustness exercises

We also conduct several robustness exercises using alternative treatment classifications. First, we vary the threshold used to define high-exposure stations by re-estimating the models using the top 40% and top 50% stations ranked by pre-program call-based reports. Second, we construct an alternative exposure measure based on in-person reporting activity, defining treatment as the top 30% stations in the number of complaints filed directly at the police station using CMS records. Finally, we implement a classification based on the police administrative typology of stations, distinguishing urban and rural police stations. This alternative definition evaluates the results by differences in urbanization or population density. Consistent results across these variants support the interpretation that effects reflect differential exposure to the VWPS reporting channel rather than arbitrary sample splits or broad differences across stations.

5.3 Heterogeneous effects

Because the VWPS changes access to law enforcement rather than policing capacity, its impact may depend on social norms governing women’s mobility and public interaction with state institutions. We further examine whether the impact of the VWPS varies with local social norms. In particular, we estimate heterogeneous treatment effects according to whether a police station is located in a socially conservative district.

We construct a measure of district conservatism using electoral outcomes from the 2018 general elections. For each district, we compute the share of votes obtained by religious political parties relative to the total number of valid votes. Districts with a vote share above the median are classified as conservative, while the remaining districts are classified as non-conservative. This classification is predetermined with respect to the program and therefore not affected by the introduction of the VWPS.

To estimate heterogeneous effects, we estimate the event-study specification separately for each subgroup. First restricting the sample to stations located in conservative districts ($Conservative_d = 1$), and then to stations located in non-conservative districts ($Conservative_d = 0$). This approach allows the entire dynamic treatment path β_k to differ across social environments without imposing parametric restrictions on how conservatism modifies treatment effects. In particular, it permits both the magnitude and timing of the response to vary across groups.

The motivation for this exercise is that the VWPS primarily reduces the social and psychological costs of contacting law enforcement. In areas where social norms impose stronger constraints on women’s mobility and public interaction with authorities, victims may face higher stigma from physically visiting a police station. In such settings, a remote reporting channel should produce larger increases in initial contact with police, while the translation of reports into formal FIRs may remain limited. Conversely, in less conservative districts, improved access may be more likely to translate into formal case registration.

6 Principal results

Figures 2 and 3 present the event-study estimates of the impact of the Virtual Women Police Station (VWPS). The horizontal axis shows time relative to the program’s introduction, and the vertical axis plots the estimated treatment effects with 95% confidence intervals. The dependent variables are expressed as year-over-year differences, defined as the change in the number of cases reported in a given month relative to the same month in the previous year. Accordingly, the coefficients do not measure changes in total reported cases, but rather deviations from the station’s expected seasonal reporting level: a positive coefficient indicates more cases than normally expected for that month, while a negative coefficient indicates fewer. Each figure also reports the pooled pre- and post-period coefficients with their standard errors, as well as the mean of the original outcome and the mean of the differenced outcome to provide a reference for the magnitude of the effects. The omitted period is

the month immediately prior to implementation, so pre-treatment coefficients test the parallel trends assumption and post-treatment coefficients trace the dynamic response to the program.

Figure 2 and Table A1 present the event-study estimates for domestic violence against women. The pre-treatment coefficients are close to zero and statistically insignificant across all outcomes, indicating that treated and control stations followed similar year-over-year trends prior to the introduction of the VWPS. This supports the parallel trends assumption. After the program's introduction, the effects diverge sharply across reporting channels.

In the walk-in reports, we observe a large and persistent negative effect on in-person reporting at police stations. The post-period coefficient is approximately -4 cases. Because the dependent variable is expressed as a year-over-year difference, this estimate implies that treated stations recorded about four fewer walk-in domestic-violence cases per month than would normally be expected for that same calendar month. Relative to an original monthly mean of roughly 3 cases, this corresponds to a substantial reduction in in-person reporting activity. Importantly, this does not necessarily indicate a reduction in underlying violence; rather, it reflects a shift away from physical reporting.

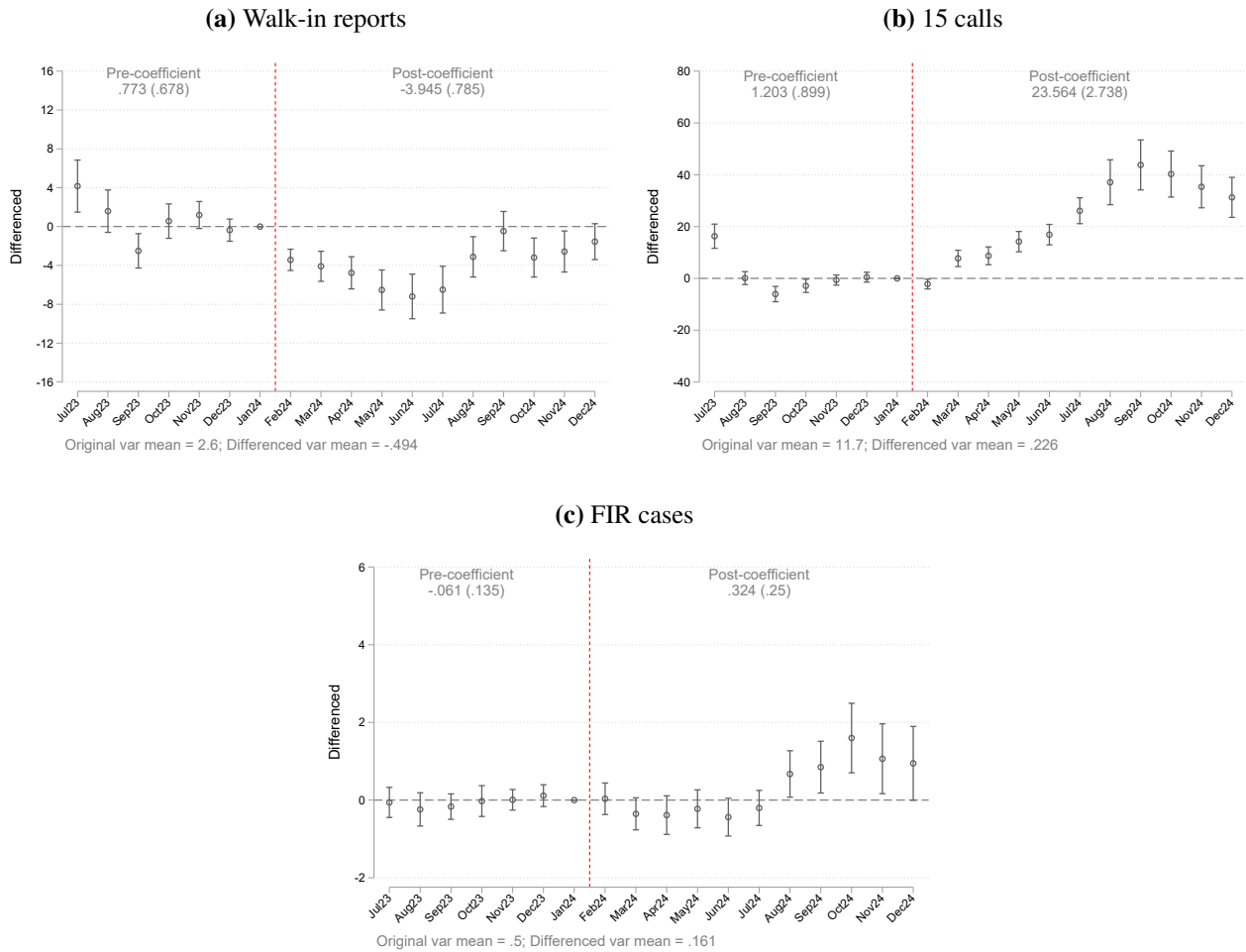
In contrast, emergency call reporting increases dramatically. The estimated effect is approximately +20 to +25 cases depending on the month. This means that treated stations received roughly twenty additional domestic-violence calls per month compared to the typical number of calls recorded in that same month one year earlier. Relative to the baseline mean (around 12 cases per month), the increase represents an expansion in reporting far exceeding normal seasonal fluctuations. The magnitude of the effect suggests that the VWPS primarily operates by facilitating initial contact with police rather than by altering crime incidence.

For FIRs, the immediate post-period effect is modest and imprecise, but becomes positive and statistically significant later in the sample. The magnitude (around +0.6 to +1 additional FIR) should be interpreted as approximately one additional formally registered domestic-violence case that became FIR per month above the expected seasonal level. Although smaller than the change in calls, this effect is large relative to the baseline FIR mean (around 0.5). This pattern indicates that the program initially affects reporting behavior, and only subsequently translates into formal legal action.

The VWPS therefore appears to reduce the fixed costs of contacting law enforcement — such as travel requirements, privacy concerns, and social stigma — especially for victims unable to physically visit a police station. Reporting domestic violence often requires leaving the household, traveling to a station, and interacting publicly with officers, which can expose victims to retaliation, reputational consequences, and family or community pressure. By enabling confidential remote contact, the program lowers these barriers to first interaction with the police. Consistent with this mechanism, we observe a substitution away from walk-in complaints toward call-based reporting and, with some delay, an increase in formal case registration (FIRs). The evidence therefore points not to an immediate reduction in underlying violence, but to improved access to the state: the VWPS makes it easier for victims to reveal incidents that would otherwise remain unreported, effectively increasing the visibility

of gender-based violence rather than directly changing its incidence.

Figure 2: Principal results - Domestic Violence



Note: Event study model estimation. The period before and after corresponds to the implementation of the program. Treated police stations are those with a high number of reported walk-in cases (top 30%). All variables are differenced from the same month of the previous year. These variables include gender-based violence cases reported directly at the police station (CMS) and by calls (Pucar), as well as the cases that progressed to a FIR. Clustered standard errors at police station level.

Appendix Figures A2–A7 provide additional evidence supporting the main interpretation and assess the robustness of the results. The pre-treatment coefficients remain centered around zero and statistically insignificant, reinforcing the absence of differential pre-trends between treated and control stations.

The program expands victims' ability to reach the police generally, lowering barriers to reporting, but the escalation of cases into formal criminal proceedings still depends on institutional procedures that are more often completed through the police station. Figure A2 supports this interpretation by extending the analysis to all gender-related cases involving female victims and the subset that progressed to an FIR. Reporting rises relative to expected seasonal levels across a broader set of

incidents, while the increase in FIRs is smaller and appears more gradually, indicating that the program affects victims' interaction with law enforcement more generally rather than only a specific crime category. Figure A3 further clarifies the mechanism by decomposing FIRs according to the initial reporting channel. Although call-based contacts increase immediately, their conversion into FIRs remains limited in the short run, whereas walk-in cases account for most of the growth in formal registration. Together, these figures point to a two-stage process in which the VWPS first facilitates initial contact with the police and only later, and imperfectly, translates these contacts into formal legal action.

The VWPS specifically affects reporting of gender-based violence rather than general police activity. Figure A4 presents placebo outcomes using crime categories unrelated to violence against women. These outcomes display no systematic post-treatment changes and no clear discontinuity around the implementation date. The absence of effects in placebo crimes indicates that the estimated impacts are not driven by general changes in police activity, reporting practices, or broader crime trends.

Appendix Figures A5–A7 assess whether the results depend on how treatment exposure is defined and show that the main conclusions are robust across alternative classifications. Figure A5 varies the cutoff used to identify high-exposure stations by defining treatment as the top 40% and top 50% of stations in baseline call-based reporting; the same qualitative pattern emerges, with a decline in walk-in complaints, a substantial rise in call-based reporting, and a delayed increase in FIRs, indicating that the findings are not driven by the specific 30% threshold. Figure A6 instead defines treatment using in-person reporting intensity, classifying as treated the top 30% of stations in walk-in complaints prior to the program; the estimated effects remain qualitatively similar but are smaller in magnitude, consistent with the program's mechanism since the VWPS primarily expands remote access to the police rather than physical access to stations. Finally, Figure A7 replaces the exposure-based treatment with the police administrative classification of urban and rural stations. Under this specification, the reallocation from walk-in reports toward calls is still present, while FIRs increase toward the end of the period in both urban and rural locations with similar dynamics (see Figure A10). The absence of differential effects across these groups suggests that the escalation of cases into formal criminal registration follows comparable institutional procedures in both settings, reinforcing the interpretation that the VWPS primarily reduces access barriers rather than changing enforcement intensity.

To better understand the source of the estimated effects, we examine the dynamics of reporting separately within treated and control stations. Rather than comparing the groups only through their difference, this exercise shows how reporting evolves over time within each group, allowing us to identify which side of the comparison drives the estimated treatment effects. Figure A8 presents the baseline classification (top 30% exposure). The results reveal that the main changes occur in treated stations: walk-in reporting falls sharply after implementation, call-based reporting increases substantially, and FIRs rise gradually over time. In contrast, control stations display relatively stable

patterns, with only small fluctuations in reporting and minimal changes in FIRs. This indicates that the estimated effects are not produced by a general time trend affecting all police stations, but by a behavioral response concentrated in high-exposure locations. Figure A9 repeats the exercise using the top 50% treatment definition and shows nearly identical dynamics, confirming that the pattern does not depend on the particular cutoff used to define treatment neither on the number of observations in each group. Finally, Figure A10 implements the same decomposition using the urban–rural classification. Both groups exhibit modest increases in FIRs toward the end of the period, but treated (urban) stations experience much larger shifts from walk-in to call-based reporting. Taken together, these figures demonstrate that the DiD estimates are driven by a clear substitution in reporting channels within treated stations rather than by common trends across all stations, reinforcing the interpretation that the VWPS primarily changes how victims contact the police rather than broader policing activity.

6.1 Heterogeneous effects by social conservatism

Figure 3 studies whether the impact of the VWPS depends on local social norms by estimating the event-study separately for police stations located in conservative and non-conservative districts. The results reveal a striking asymmetry in how the program translates into reporting and legal action.

In conservative districts, the introduction of the VWPS produces a large and immediate increase in call-based reporting together with a pronounced decline in walk-in complaints, yet the response in FIRs remains limited. The natural interpretation is that the program relaxes the most binding constraint victims face in these settings: the ability to contact the police without public exposure. In socially restrictive environments, reporting violence often entails leaving the household, being seen entering a police station, and potentially confronting family or community members who oppose formal reporting. The virtual channel allows victims to communicate privately with authorities, making it possible to disclose incidents that previously remained hidden. However, the same social pressures that prevented in-person reporting also make formal legal escalation costly, so many cases stop at the stage of initial contact.

In non-conservative districts, the pattern differs. The increase in call-based reporting is smaller, but FIRs rise more quickly and more strongly. This suggests that victims in these areas were not primarily constrained by the ability to reach the police, but by coordination and procedural frictions in initiating a case. Once the VWPS facilitates contact, victims are more likely to proceed to formal case registration.

These results are consistent with a two-stage model of reporting. The first stage is contact with the state, which depends on privacy, mobility, and stigma costs; the second stage is legal escalation, which depends on social support and willingness to pursue formal action. The VWPS directly reduces the first-stage cost everywhere, but the second stage remains governed by local norms. Consequently, the program has its largest effect on initial reporting precisely where women face the greatest social barriers, while formal legal action responds more in environments where pursuing a case is socially

and institutionally feasible.

The broader implication is that the program does not simply increase reporting; it reveals previously latent violence. In conservative areas, the VWPS primarily uncovers hidden victimization by enabling confidential communication with the police. In less conservative areas, it also changes legal outcomes by allowing reported incidents to progress into formal criminal cases. The heterogeneity therefore provides direct evidence that access to justice policies interact with social norms: improving access to the state is sufficient to change reporting behavior, but changing legal enforcement requires overcoming local social constraints.

7 Conclusions

This paper studies whether reducing the cost of contacting law enforcement can expand access to justice for victims of gender-based violence. Using administrative police data and the introduction of the Women’s Virtual Police Station (VWPS) in Punjab, Pakistan, we show that a digital reporting platform substantially changes how citizens interact with the police.

The reform produces a large increase in remote reporting together with a decline in in-person complaints, indicating that victims substitute reporting channels rather than immediately experiencing changes in victimization. Formal case registration rises only gradually. This pattern reveals a sequential justice process: the main constraint is not necessarily investigation or prosecution, but the ability of victims to initiate contact with the state. The VWPS primarily affects this first stage by lowering privacy, mobility, and stigma costs associated with reporting.

The heterogeneity analysis clarifies the mechanism. In more socially conservative districts, the program generates large increases in reporting but comparatively small changes in formal case registration. In less conservative districts, a greater share of reports progresses into FIRs. These findings suggest that the reform relaxes access constraints everywhere, but the decision to pursue legal escalation depends on local social norms and the social costs of public legal action. The program therefore reveals previously hidden violence in some places and facilitates legal enforcement in others.

The results have two broader implications. First, increases in recorded crime following access-expanding reforms should not be interpreted as worsening safety. Instead, they may reflect improved reporting and the revelation of previously latent victimization. Administrative crime statistics depend not only on the prevalence of crime but also on the accessibility of the justice system. Second, improving public protection does not always require expanding police resources. Policies that reduce the cost of interacting with the state—particularly those that improve privacy and confidentiality—can significantly expand the reach of legal institutions even without changes in enforcement capacity.

More generally, the findings highlight that failures of justice systems often occur before formal legal processes begin. In many settings, victims never enter the system at all. The VWPS changes this margin by allowing individuals to reach the state without first overcoming substantial social and logistical barriers. Digital platforms therefore have the potential to transform state capacity not by

increasing coercive power, but by making public institutions reachable.

At the same time, the results also point to the limits of access reforms. While the program facilitates contact with authorities, the progression of cases into formal legal action remains mediated by institutional procedures and social constraints. Expanding reporting is a necessary but not sufficient condition for effective protection. Future policies may need to complement access-oriented reforms with interventions that address investigation, victim support, and social barriers to prosecution.

Taken together, the evidence suggests that the primary role of the VWPS is not to reduce violence directly, but to reduce the invisibility of violence. By lowering the cost of contacting the police, the program makes harm observable to the state. In contexts where legal protection depends on being seen and heard, that first interaction may be the most important step of all.

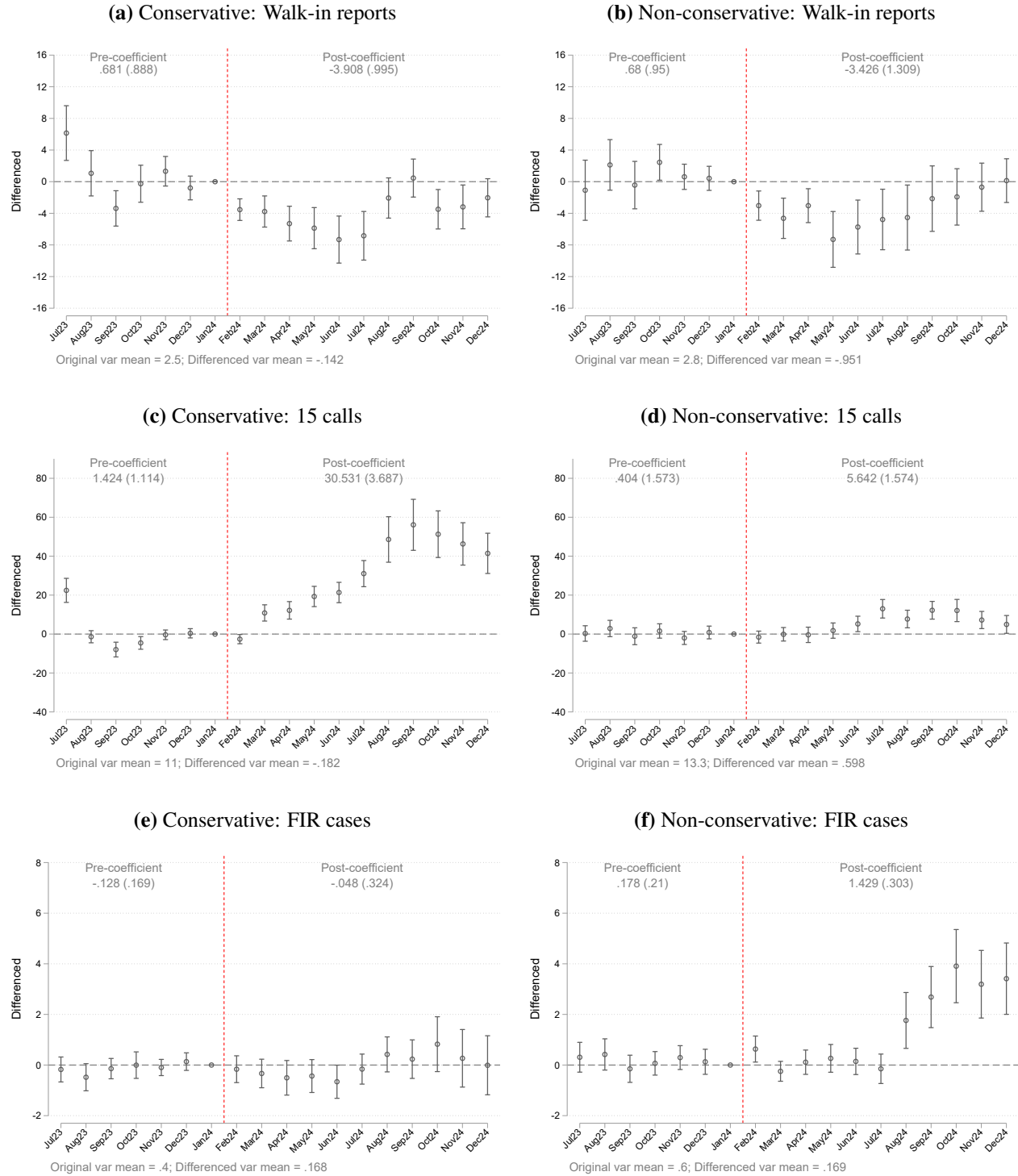
Digital Access to Justice: Evidence from Women's Virtual Police Stations in Pakistan*

Bakhtawar Ali Saad Gulzar Muhammad Yasir Khan Muhammad Ahsan
Yonus Amna Baig

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Figure 3: Domestic Violence by Conservative and Non-conservative districts



Note: Event study model estimation. The period before and after corresponds to the implementation of the program. Treated police stations are those with a high number of reported walk-in cases (top 30%) in conservative (higher proportion of votes for religious parties in 2018) and non-conservative districts. All variables are differentiated from the same month of the previous year. These variables include gender-based violence cases reported directly at the police station (CMS) and by calls (Pucar), as well as the cases that progressed to a FIR. Clustered standard errors at police station level.

References

- Acemoglu, D. and Jackson, M. O. (2017) Social norms and the enforcement of laws, *Journal of the European Economic Association*, **15**, 245–295.
- Aizer, A. and Dal Bo, P. (2009) Love, hate and murder: Commitment devices in violent relationships, *Journal of public Economics*, **93**, 412–428.
- Amaral, S., Bhalotra, S. and Prakash, N. (2021) Gender, crime and punishment: Evidence from women police stations in india, *Journal of Development Economics*.
- Amaral, S., Borker, G., Fiala, N., Kumar, A., Prakash, N. and Sviatschi, M. M. (2023) Sexual harassment in public spaces and police patrols: Experimental evidence from urban india, Working Paper 31734, National Bureau of Economic Research.
- Banerjee, A., Duflo, E., Keniston, D. and Singh, N. (2021) Improving police performance in rajasthan, india: Experimental evidence on incentives, managerial autonomy, and training, *American Economic Journal: Economic Policy*, **13**, 1–40.
- Blattman, C., Jamison, J. C. and Sheridan, M. (2017) Reducing crime and violence: Experimental evidence from cognitive behavioral therapy in liberia, *American Economic Review*, **107**, 1165–1206.
- Callen, M., Gulzar, S., Hasanain, A., Khan, M. Y. and Rezaee, A. (2020) Data and policy decisions: Experimental evidence from pakistan, *Journal of Development Economics*, **146**, 102486.
- Cheema, A., Khan, A. and Myerson, R. (2015) Breaking the countercyclical pattern of local democracy in pakistan, *Is decentralization good for development*, pp. 68–79.
- Cheema, A., Khan, A. Q. and Myerson, R. B. (2014) Breaking the countercyclical pattern of local democracy in pakistan, Tech. rep., International Growth Centre Working Paper.
- Chin, Y.-M. and Cunningham, S. (2019) Revisiting the effect of warrantless domestic violence arrest laws on intimate partner homicides, *Journal of Public Economics*, **179**, 104072.
- Clots-Figueras, I., Bhalotra, S. and Iyer, L. (2017) Pathbreakers? women’s electoral success and future political participation.
- Draca, M., Machin, S. and Van Reenen, J. (2011) Panic on the streets of london: Police, crime, and the july 2005 terror attacks, *American Economic Review*, **101**, 2157–2181.
- Gangadharan, L., Jain, T., Maitra, P. and Vecchi, J. (2019) Female leaders and their response to the social environment, *Journal of Economic Behavior & Organization*, **164**, 256–272.

- Hessami, Z. and Da Fonseca, M. L. (2020) Female political representation and substantive effects on policies: A literature review, *European Journal of Political Economy*, **63**, 101896.
- Iyer, L., Mani, A., Mishra, P. and Topalova, P. (2012) The power of political voice: women's political representation and crime in india, *American Economic Journal: Applied Economics*, **4**, 165–193.
- Jassal, N. (2020) Gender, law enforcement, and access to justice: Evidence from all-women police stations in india, *American Political Science Review*, **114**, 1035–1054.
- Jassal, N. (2024) Does victim gender matter for justice delivery? police and judicial responses to women's cases in india, *American Political Science Review*.
- Jensen, R. and Oster, E. (2009) The power of tv: Cable television and women's status in india, *The Quarterly Journal of Economics*, **124**, 1057–1094.
- Kavanaugh, G., Sviatschi, M. M. and Trako, I. (2019) Gender violence, enforcement, and human capital: Evidence from all-women's justice centers in peru, *Available at SSRN*, **3022670**, 38.
- Kumar, S. and Prakash, N. (2017) Effect of political decentralization and female leadership on institutional births and child mortality in rural bihar, india, *Social Science & Medicine*, **185**, 171–178.
- Lewis-Faupel, S., Neggers, Y., Olken, B. A. and Pande, R. (2016) Can electronic procurement improve infrastructure provision? evidence from public works in india and indonesia, *American Economic Journal: Economic Policy*, **8**, 258–283.
- Mello, S. (2019) “they can't search her”: How gender imbalances in the police force contribute to perceptions of procedural unfairness, *Social Problems*.
- Miller, A. R. and Segal, C. (2019a) Do female officers improve law enforcement quality? effects on crime reporting and domestic violence, *The Review of Economic Studies*, **86**, 2220–2247.
- Miller, A. R. and Segal, C. (2019b) Do female officers improve law enforcement quality? effects on crime reporting and domestic violence, *The Review of Economic Studies*, **86**, 2220–2247.
- Olken, B. A. (2007) Monitoring corruption: Evidence from a field experiment in indonesia, *Journal of Political Economy*, **115**, 200–249.
- Perova, E. and Reynolds, S. (2017) Women's police stations and intimate partner violence: Evidence from brazil, *Social Science & Medicine*, **174**, 188–196.
- Sukhtankar, S., Kruks-Wisner, G. and Mangla, A. (2022a) Policing in patriarchy: An experimental evaluation of reforms to improve police responsiveness to women in india, *Science*, **377**, 191–198.

Sukhtankar, S., Kruks-Wisner, G. and Mangla, A. (2022b) Policing in patriarchy: An experimental evaluation of reforms to improve police responsiveness to women in india, *Science*, **377**, 191–198.

8 Appendix

Figure A1: Gender cases classification

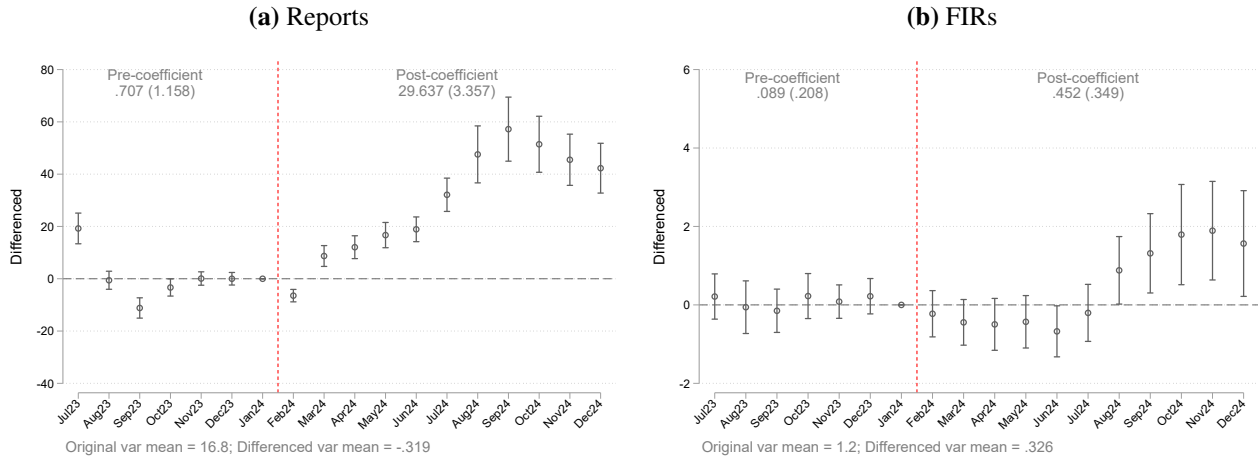
Case	CMS	Pucar
Domestic violence	Violence Against Woman	Domestic Violence
Other gender cases	Acid Throwing Domestic Violence Gang Rape Harassment at workplace Kidnapping Minors Outraging the Modesty of Women Rape Unlawful Marriages	Acid Throwing Female Kidnapping/ Abduction Physical Threats / Harrasment Prostitution/ Brothel House Rape Sexual Assault/ Harrasment To Women

Table A1: Principal results

Dep. Variable	Reports		FIRs
	(Walk-in) (1)	(15 calls) (2)	(3)
Urban×Aug23	1.593 (1.133)	0.016 (1.293)	-0.290 (0.224)
Urban×Sep23	-2.465*** (0.916)	-6.163*** (1.522)	-0.195 (0.172)
Urban×Oct23	0.659 (0.909)	-3.009** (1.309)	-0.057 (0.207)
Urban×Nov23	1.162 (0.712)	-0.780 (1.022)	-0.006 (0.140)
Urban×Dec23	-0.435 (0.589)	0.211 (0.991)	0.098 (0.146)
Urban×Feb24	-3.198*** (0.567)	-2.031** (0.974)	0.013 (0.210)
Urban×Mar24	-3.912*** (0.801)	7.722*** (1.615)	-0.330 (0.213)
Urban×Apr24	-4.521*** (0.859)	8.658*** (1.758)	-0.348 (0.257)
Urban×May24	-6.247*** (1.067)	13.648*** (2.012)	-0.197 (0.254)
Urban×Jun24	-6.995*** (1.186)	16.179*** (2.022)	-0.428* (0.251)
Urban×Jul24	-6.297*** (1.243)	24.574*** (2.559)	-0.201 (0.235)
Urban×Aug24	-2.879*** (1.080)	36.184*** (4.426)	0.605* (0.312)
Urban×Sep24	-0.329 (1.052)	42.809*** (4.925)	0.719** (0.348)
Urban×Oct24	-3.160*** (1.037)	39.467*** (4.533)	1.498*** (0.462)
Urban×Nov24	-2.522** (1.091)	34.808*** (4.139)	0.968** (0.463)
Urban×Dec24	-1.409 (0.956)	30.893*** (3.939)	0.873* (0.488)
<i>Linear combinations P-values</i>			
Pre treatment	0.874	0.047	0.503
Post treatment	0.000	0.000	0.256
Mean of original variable variable	02.8	13.2	00.5
Mean of differenced variable	-0.646	0.368	0.193
Number of obs.	12512	12512	12512
Number of Police Stations	736	736	736
Police Station FE	Yes	Yes	Yes
Police Station Cluster	Yes	Yes	Yes

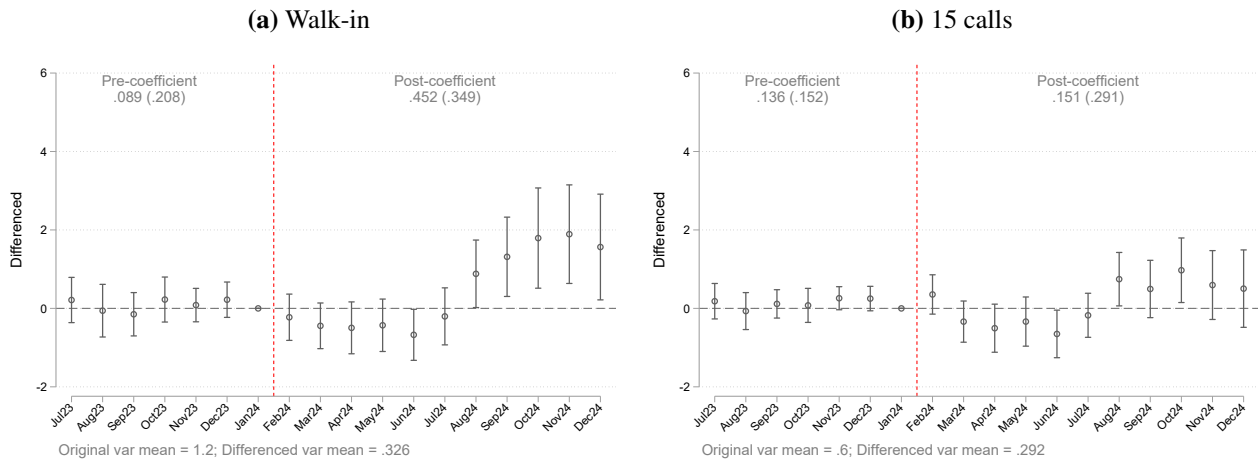
Notes: Event study model estimation. The period before and after corresponds to the implementation of the program. Treated police stations are those with a high number of reported walk-in cases (top 30%). All variables are differentiated from the same month of the previous year. These variables include gender-based violence cases reported directly at the police station (CMS) and by calls (Pucar), as well as the cases that progressed to a FIR. Clustered standard errors at police station level.

Figure A2: All gender cases report



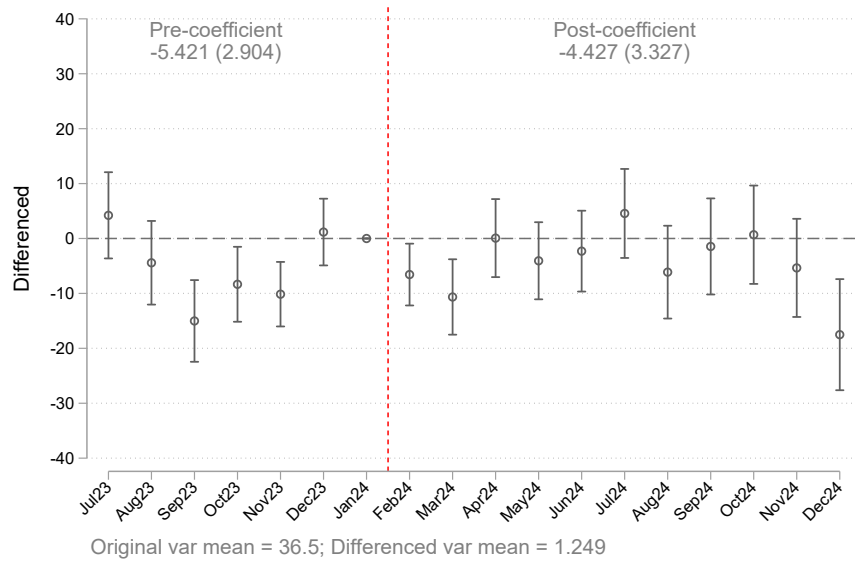
Note: Event study model estimation. The period before and after corresponds to the implementation of the program. Treated police stations are those with a high number of reported walk-in cases (top 30%). All variables are differenced from the same month of the previous year. These variables include all gender violence cases reported by both calls (Pucar) and directly at the police station (CMS), and cases that progressed to a FIR. Clustered standard errors at police station level.

Figure A3: FIRs cases by way of report



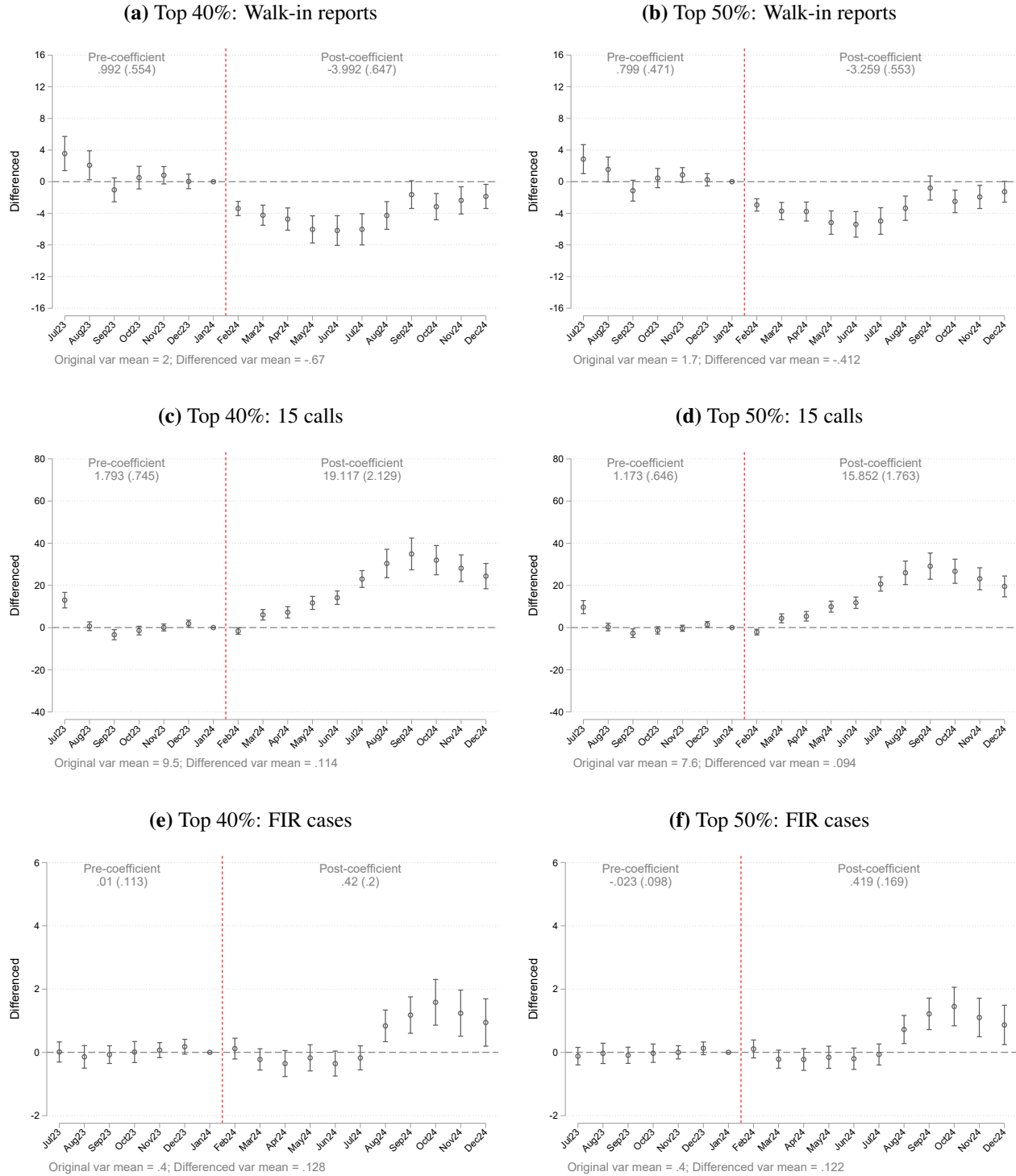
Note: Event study model estimation. The period before and after corresponds to the implementation of the program. Treated police stations are those with a high number of reported walk-in cases (top 30%). All variables are differenced from the same month of the previous year. These variables include all gender violence cases that progressed to a FIR split by the way of report calls (Pucar) and directly at the police station (CMS). Clustered standard errors at police station level.

Figure A4: Other type of crime reporting



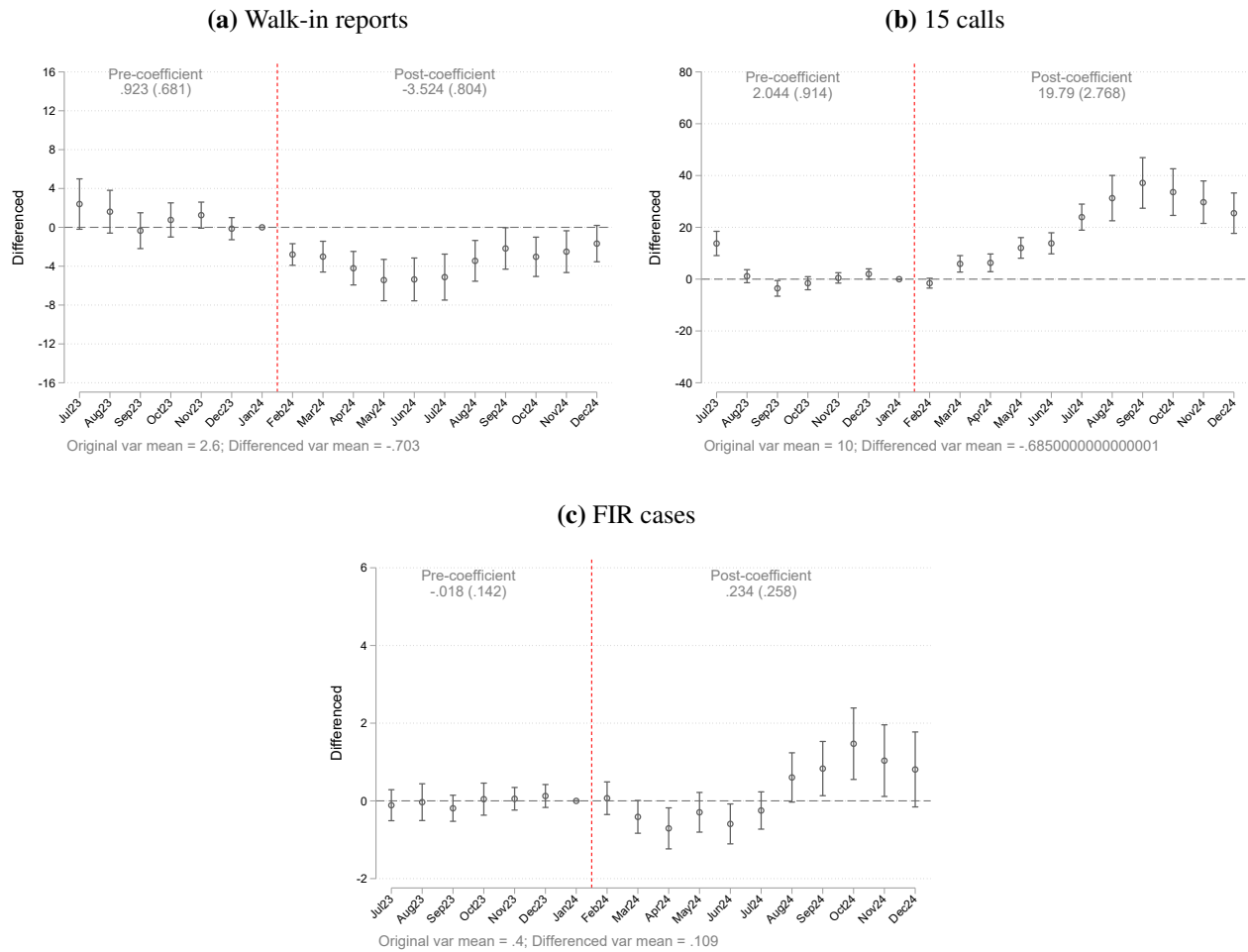
Note: Event study model estimation. The period before and after corresponds to the implementation of the program. Treated police stations are those with a high number of reported walk-in cases (top 30%). All variables are differentiated from the same month of the previous year. These variables include other type of crime not related to gender cases reported directly at the police station (CMS). Clustered standard errors at police station level.

Figure A5: Robustness results - Top 40% and 50%



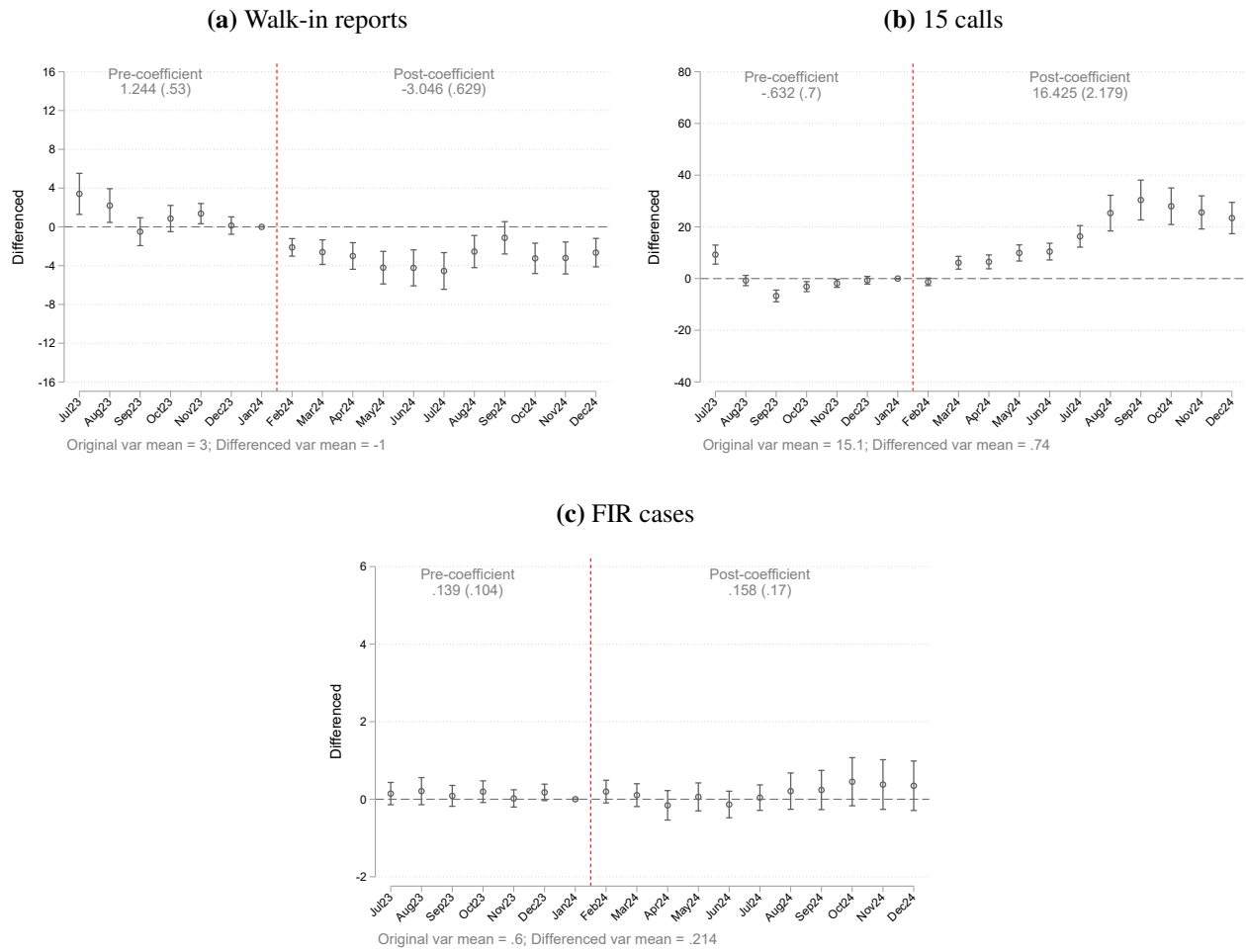
Note: Event study model estimation. The period before and after corresponds to the implementation of the program. Treated police stations are those with a high number of reported walk-in cases: top 40% and top 50%. All variables are differentiated from the same month of the previous year. These variables include gender-based violence cases reported directly at the police station (CMS) and by calls (Pucar), as well as the cases that progressed to a FIR. Clustered standard errors at police station level.

Figure A6: Robustness results - Top 30% of walk-in cases



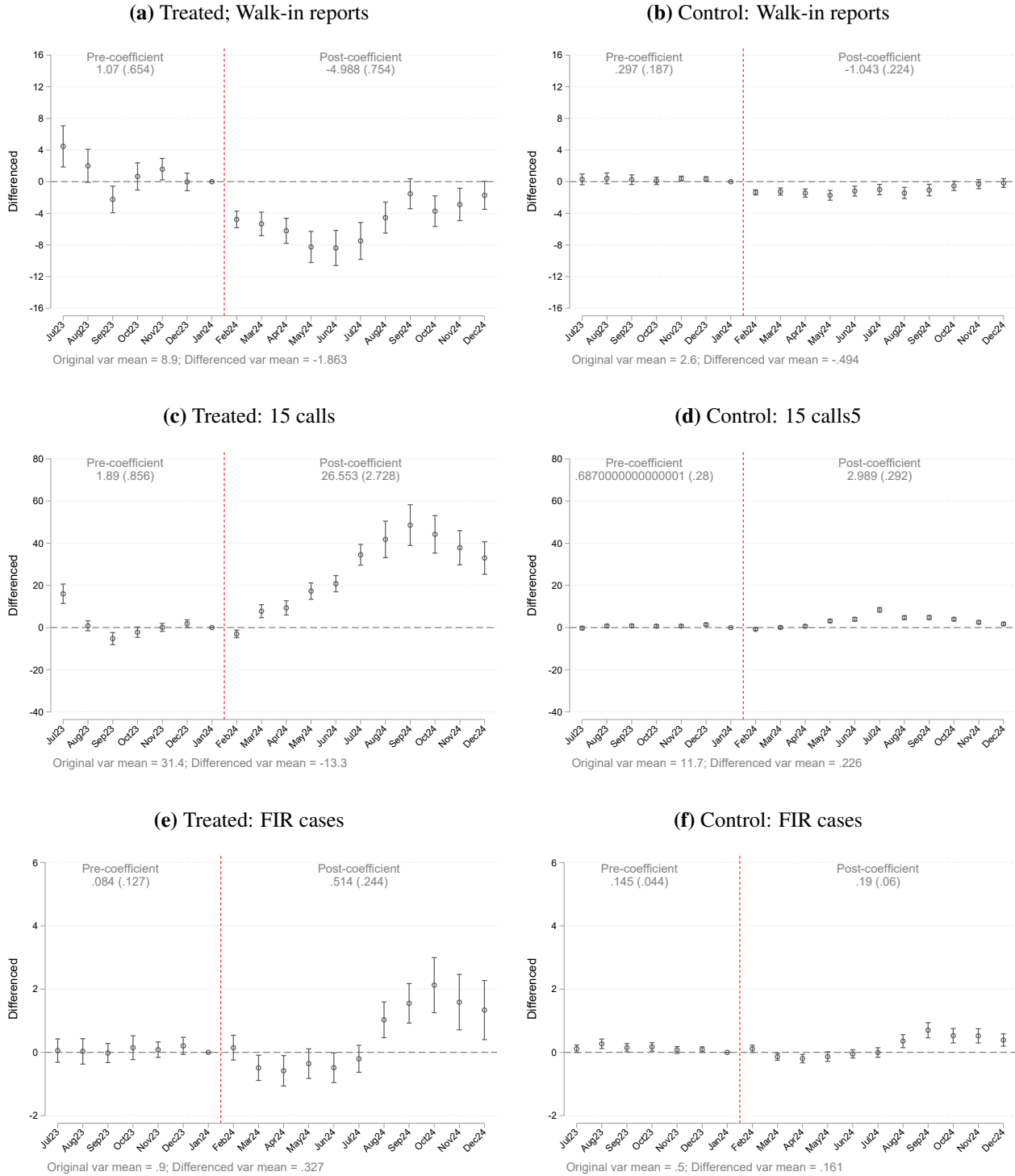
Note: Event study model estimation. The period before and after corresponds to the implementation of the program. Treated police stations are those with a high number of reported by 15 calls (top 30%). All variables are differentiated from the same month of the previous year. These variables include gender-based violence cases reported directly at the police station (CMS) and by calls (Pucar), as well as the cases that progressed to a FIR. Clustered standard errors at police station level.

Figure A7: Robustness results - Urban Police clasification



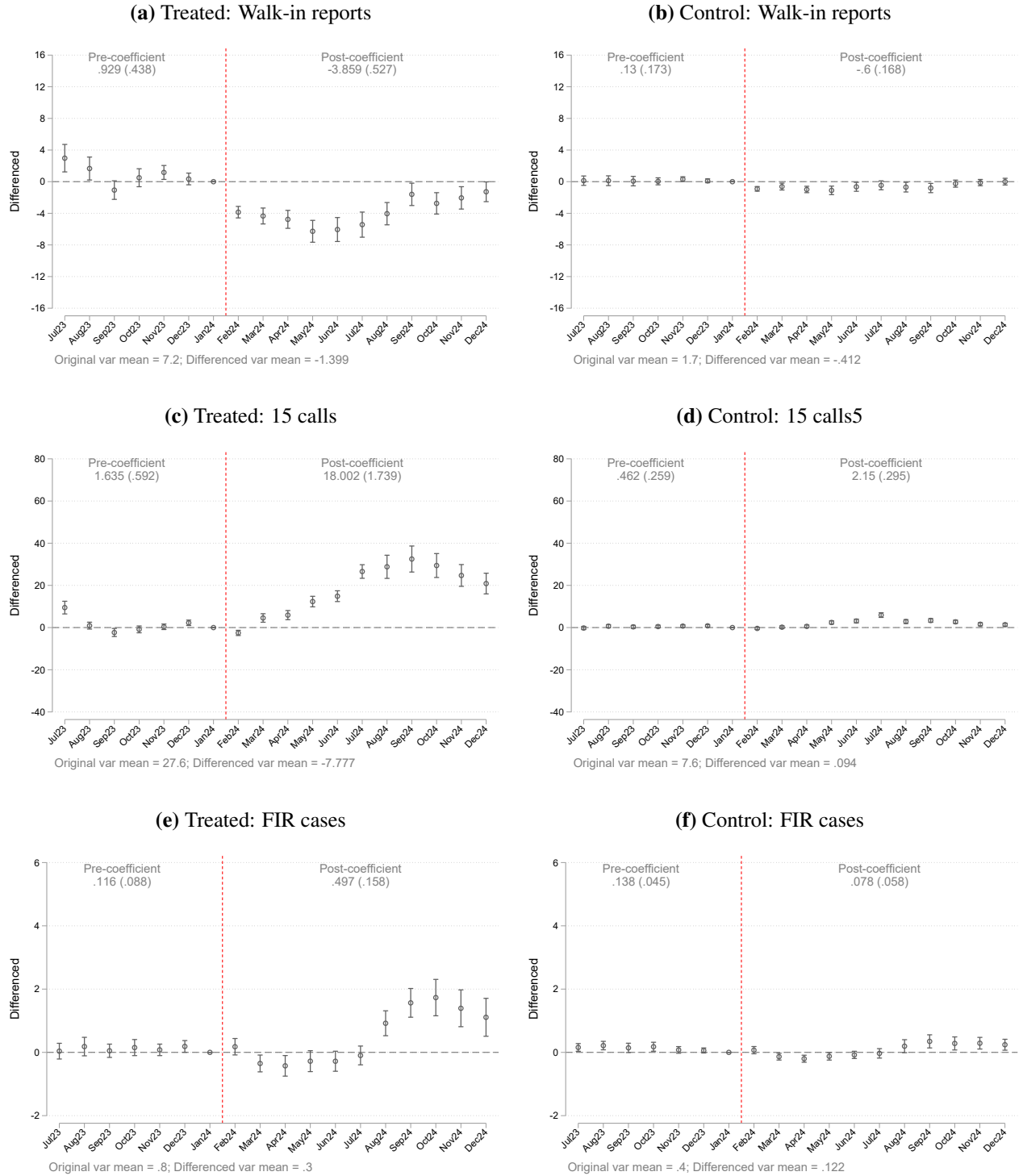
Note: Event study model estimation. The period before and after corresponds to the implementation of the program. Treated police stations are those categorized as urban and semi-urban from the Police classification. All variables are differentiated from the same month of the previous year. These variables include gender-based violence cases reported directly at the police station (CMS) and by calls (Pucar), as well as the cases that progressed to a FIR. Clustered standard errors at police station level.

Figure A8: Treated and control comparison over time



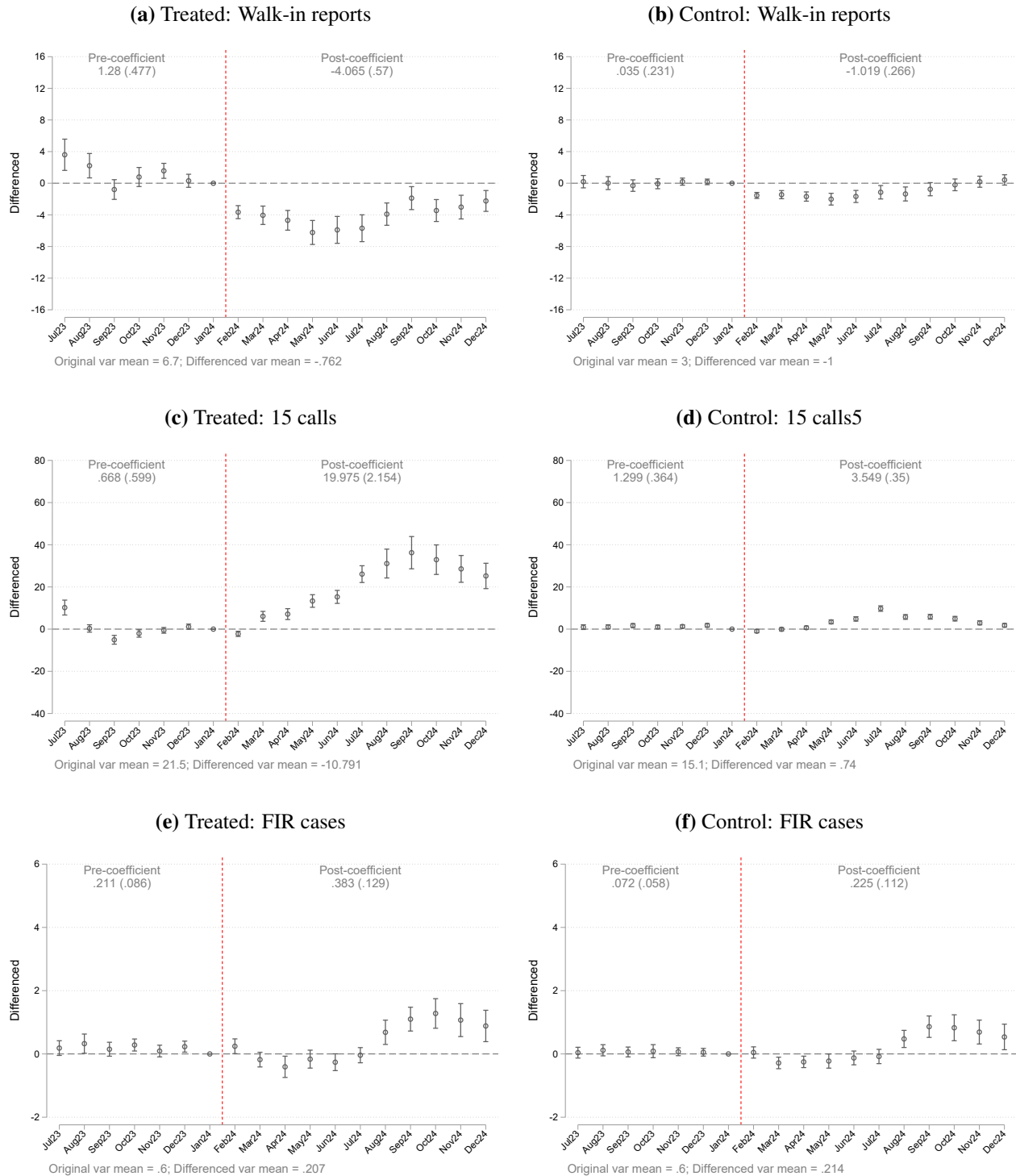
Note: Event study model estimation only for treated and only for control units. The period before and after corresponds to the implementation of the program. Treated police stations are those with a high number of reported walk-in cases (top 30%). All variables are differentiated from the same month of the previous year. These variables include gender-based violence cases reported by calls (Pucar) and directly at the police station (CMS), and cases that progressed to a FIR. Clustered standard errors at police station level.

Figure A9: Treated and control comparison over time - Top 50%



Note: Event study model estimation only for treated and only for control units. The period before and after corresponds to the implementation of the program. Treated police stations are those with a high number of reported walk-in cases (top 50%). All variables are differentiated from the same month of the previous year. These variables include gender-based violence cases reported by calls (Pucar) and directly at the police station (CMS), and cases that progressed to a FIR. Clustered standard errors at police station level.

Figure A10: Treated and control comparison over time - Urban/rural classification



Note: Event study model estimation only for treated and only for control units. The period before and after corresponds to the implementation of the program. Treated police stations are those located in urban areas. All variables are differenced from the same month of the previous year. These variables include gender-based violence cases reported by calls (Pucar) and directly at the police station (CMS), and cases that progressed to a FIR. Clustered standard errors at police station level.